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ITAI 1371 Intro to Machine Learning

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L10 Reflection Journal

I've always worked with supervised learning, so in this lab I learned that the key detail in working with unsupervised learning was double-checking the end result. In earlier labs, I was always able to check a model against the ground truth, but K-Means has none. The elbow plot showed me that, because inertia keeps going down as k goes up, so the best k comes from where the curve bends and not from the lowest number. I also noticed that make_blobs creates labels that we never used in this lab, and that also showed me that K-Means finds the 4 groups strictly from the data shape.

PCA connects back to Lab 02, where I plotted the Iris features and saw setosa form its own group while versicolor and virginica overlapped. This time, PCA put all 4 measurements into 2 components that keep 97.77% of the variance, and I noticed that same pattern in the plot. This means that the 4 features share the majority of their information. It also made me ask why the lab skips the scaling step that the slides require before PCA. It works here because all 4 features use centimeters, but our Telco dataset for the final mixes months and dollars, so I would likely need to standardize it first.

Reflective Journal

ITAI 1371 | Lab 10: Unsupervised Learning | Student: Elquin Ponce

Before this lab, I thought of machine learning as something that always needed an answer key. Every model we built earlier had labels, so the idea of a model finding structure without them felt backwards to me. What made it click was the first scatter plot of the synthetic customer data. I could see four loose groups by eye, but the algorithm had never been told they were there. K-Means had to find them using only distance, which showed me that "unsupervised" means there is no right answer to check against, only patterns the data happens to contain.

The elbow method and why bigger k is not better

The elbow method was the part I understood least at first. I could not see why we would not just pick the largest k, since inertia keeps falling as k grows. Plotting it made the reason clear. Once every point is its own cluster, inertia reaches zero, but the clusters no longer mean anything. The useful k is where the curve bends and extra clusters stop adding much. My plot bent at k = 4, which matched the four blobs I could see in the raw data. That also showed me the elbow is a judgment call. This data had a clear bend, and I still wonder how I would choose k on real data where the bend is not so obvious.

PCA and what 97.77% tells me

PCA was the other big idea. Reducing the Iris dataset from four features to two kept 97.77% of the variance, with the first component alone carrying about 92%. To me that means the four measurements are heavily redundant, so most of the information can be described with far fewer dimensions. The 2D plot also separated the three species, even though PCA never saw the species labels. One species formed its own group, while the other two overlapped somewhat. Seeing structure appear without labels made the idea of unsupervised learning much more concrete than reading the definition did.

A challenge I had to debug

My biggest struggle was technical. The first code cell failed with a SyntaxError because the notebook file had lost its line breaks, so several statements were squashed onto one line. At first I assumed I had made a mistake in the code. I worked it out by reading the error closely, noticing that two statements were joined together, and rewriting the cell with proper line breaks. I also found that the Task 3 plotting code used seaborn without importing it, so I added the import. This taught me that reading an error message carefully is a real skill, and that not every error is my own mistake.

Connections to earlier work and what I still wonder

These ideas connect to my midterm work on scaling and normalization. K-Means and PCA both depend on distance and variance, so features on very different scales could distort the results. This lab used the raw Iris data, and I now suspect that scaling it first would change the variance percentages. I would like to test that. I also want to understand what exactly is lost when components are dropped, and how that tradeoff changes when the data is not as redundant as Iris.

Overall, the workflow no longer feels like a list of steps to run. Each step answers a question: how many groups are there, and how much of the data can I keep while simplifying it. That is the main thing I will take from this lab.

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Reflection Journal

In Module 10 Lab on Unsupervised Learning, I explored patterns and underlying structures in unlabeled datasets without pre-assigned ground truth target variables. In previous labs, I worked primarily with supervised learning models using labeled data. The primary objective was to implement clustering algorithms, specifically *K*-Means and Hierarchical Clustering—alongside techniques like Principal Component Analysis (PCA. Working with unsupervised algorithms shifted my analytical mindset from predictive mapping (input-to-output target matching) to exploratory discovery. Determining optimal cluster counts (*K*) required using diagnostic heuristics, such as the Elbow Method.

In task 1, We use the elbow method to find the optimal number of clusters. The number turned out to be 4. With k=4, I ran K-Means to create the scatterplot, displaying the clusters with assigned colors. Moving onto task 3, we proceed to use Principal Component Analysis (PCA)concept on the Iris data to reduce the features from 4 to 2, and plot onto a scatterplot. Principal Component 1 explains 92.46% of the variance while Principal Component 2: Explains 5.31% of the variance. The two components combined explain 97.77% of the total dataset variance, confirming high information retention while successfully reducing dimensions for 2D visualization.

After working on this lab, I have learned that unsupervised learning techniques provide essential tools for exploratory data analysis and anomaly detection. Moving forward, I plan to integrate PCA preprocessing pipelines with supervised classification models to reduce noise and optimize feature selection in high-dimensional engineering datasets.

Module 10 Lab - Unsupervised Learning Reflection Journal

Knowledge Check Questions:

1. **What is the fundamental difference between supervised and unsupervised learning?**
    a. Supervised learning uses labeled data (X and Y) to guide model training, while unsupervised learning uses unlabeled data (X only), seeking to identify any underlying structures, patterns, or groupings within the data without any target labels.

2. **In the K-Means elbow plot, why don't we just choose the largest possible k to get the lowest inertia?**
    a. Choosing the largest possible k would result in an inertia of 0. This causes overfitting of the data and defeats the purpose of clustering. Using the elbow method helps to find the point of diminishing returns.

3. **In the PCA results, the first two principal components explained over 95% of the variance. What does this tell you about the original 4-dimensional Iris dataset?**
    a. This tells us that the original 4D dataset contains redundant information. Because more than 97.77% of the dataset's total variance can be captured in just 2 dimensions, we can confidently compress and visualize the dataset in a 2D space while preserving almost all of its original information.

Through this lab, I explored unsupervised learning by grouping synthetic customer data with K-Means clustering and reducing the feature space of the Iris dataset using PCA. By implementing the Elbow Method, I learned to determine the optimal cluster count mathematically, identifying four distinct customer segments. Additionally, applying PCA demonstrated the power of dimensionality reduction, allowing us to compress four flower features into two principal components while retaining 97.77% of the dataset's total variance. Overall, this experience showed me how unsupervised techniques can discover hidden structures and make complex data easy to analyze.

Kei'Chara Fleeks

ITAI1371_L10 Reflective Journal_ Diane Lenhoff

What I Learned

In this lab, I learned the difference between supervised and unsupervised learning. With supervised learning, the model uses labeled data to make predictions, while unsupervised learning looks for patterns without knowing the correct answers. I also learned how K-Means clustering groups similar data points and how the elbow method helps determine the best number of clusters. In our exercise, four clusters appeared to be the best choice.

Challenges

The most challenging part for me was understanding how the elbow method determines the number of clusters and why we would not just choose the largest number to get the lowest inertia. I also had to spend some time understanding PCA and how reducing the number of features could still preserve most of the information in the dataset.

What Surprised Me

I was surprised that PCA could reduce the Iris dataset from four features to two principal components while still explaining 97.77% of the variance. I expected that reducing the number of dimensions would result in losing more information. This helped me understand how PCA can simplify data while keeping most of the important information.

Overall Reflection

This lab helped me better understand how machine learning can identify patterns without labeled data. I can see how clustering could be useful in business for identifying customer groups based on purchasing habits. PCA could also be useful when working with large datasets because it can reduce the number of features and make the data easier to analyze.